Obviously, not everything is digital, like music. Can we reduce the cost of non-digital goods and services too, like housing and clothes? Can we create their Spotify equivalents, "Constructify" and "Clothify"?

We can, if we do three things. First, we must take the expensive humans out of the production process for all goods and services. That is exactly what automation does, so this is a case where the problem is also part of the cure.

Second, we must make energy very cheap. The cost of solar cells is falling fast, and battery technology is improving. Many observers think that within 20 or 30 years, electricity could not only be much cheaper to generate, store and transmit than the oil and gas we get by digging up dead dinosaurs – it could be almost too cheap to meter. This would also reduce CO2 emissions much faster than most people today think possible.

Third, we must use AI to make all production processes as efficient as possible, requiring minimal material inputs.

AUTOMATE AND REDEPLOY

Andrew McAfee pointed out in his recent book "More From Less" that we are already on this path, but there is much further to go. We should therefore not hold automation back; instead we should accelerate it. The mantra for companies and for countries should be: "automate and redeploy, rinse and repeat". Companies which automate rapidly will prosper, but the outstanding ones will be those which also genuinely value their people: companies where people actively seek to work out how to automate their tasks, in the knowledge that they will be rewarded both with pay and with more interesting work in the future.

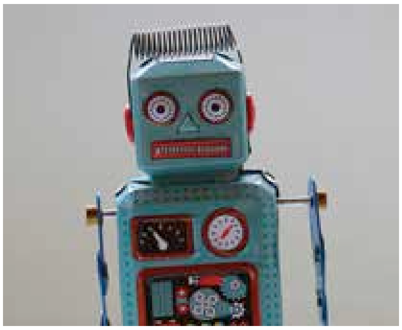
THE CHURN

For a decade or three before technological unemployment arrives, AI-driven automation will have an increasingly disruptive effect on the job market: the Churn. People will lose their jobs to machines, and they will have to be redeployed within companies, between companies, and between industries. With increasing frequency. Workers will have to learn how to work with machines: AI won't replace humans in all jobs, but increasingly, humans who can work with AI will replace those who cannot. We cannot yet say what jobs people will be re-trained to do; no doubt some of them will surprise us.

This process will be disturbing and frightening for many of us. Governments will need to support their people through the Churn with enhanced welfare programmes and other methods of economic support. The experience of massive government intervention during the corona virus crisis should teach us valuable lessons about what works and what doesn't.

EDUCATION WILL BECOME VACATIONAL, NOT VOCATIONAL

During the period of Churn, we will need to re-train ourselves more and more frequently, and do it faster each time. Education and training are notoriously hard industries to reform. Here again, AI will provide solutions as well as raising the challenge in the first place. Education and training will become personalised, as we all acquire learning assistants - digital coaches which know better than we do (and better than any human teacher could do) what we already know, what we need to learn next, and how to optimise that learning process. For many years, these assistants will work alongside human teachers rather than replacing them. If we manage the transition smoothly, then by the time the replacement happens, we will hardly notice, and we won't object. Once we arrive at the economy of abundance, some people will continue to re-train for economic activity, but for most of us, educational will become vacational, not vocational. Which is what many of us wanted it to be all along.



PANIC OR PLAN

To survive the economic singularity, we need to identify the optimal outcome in a largely post-jobs economy. If it is indeed the economy of abundance, and fully automated luxury capitalism, then we need to build a consensus for that, and a plan for how to get there without panic. Panic at the thought of racing toward technological unemployment without a plan could be almost as bad as the fact itself. But if we can manage the transition through the economic singularity successfully, the outcome will be wonderful. d

About the author:

Calum is a sought-after keynote speaker and best-selling writer on artificial intelligence. His non-fiction books on AI are Surviving AI, about strong AI and superintelligence, and The Economic Singularity, about the prospect of widespread technological unemployment. Before becoming a full-time writer and speaker, Calum had a 30-year career in journalism and in business, as a marketer, a strategy consultant and a CEO. He studied philosophy at Oxford University, which confirmed his suspicion that science fiction is actually philosophy in fancy dress.